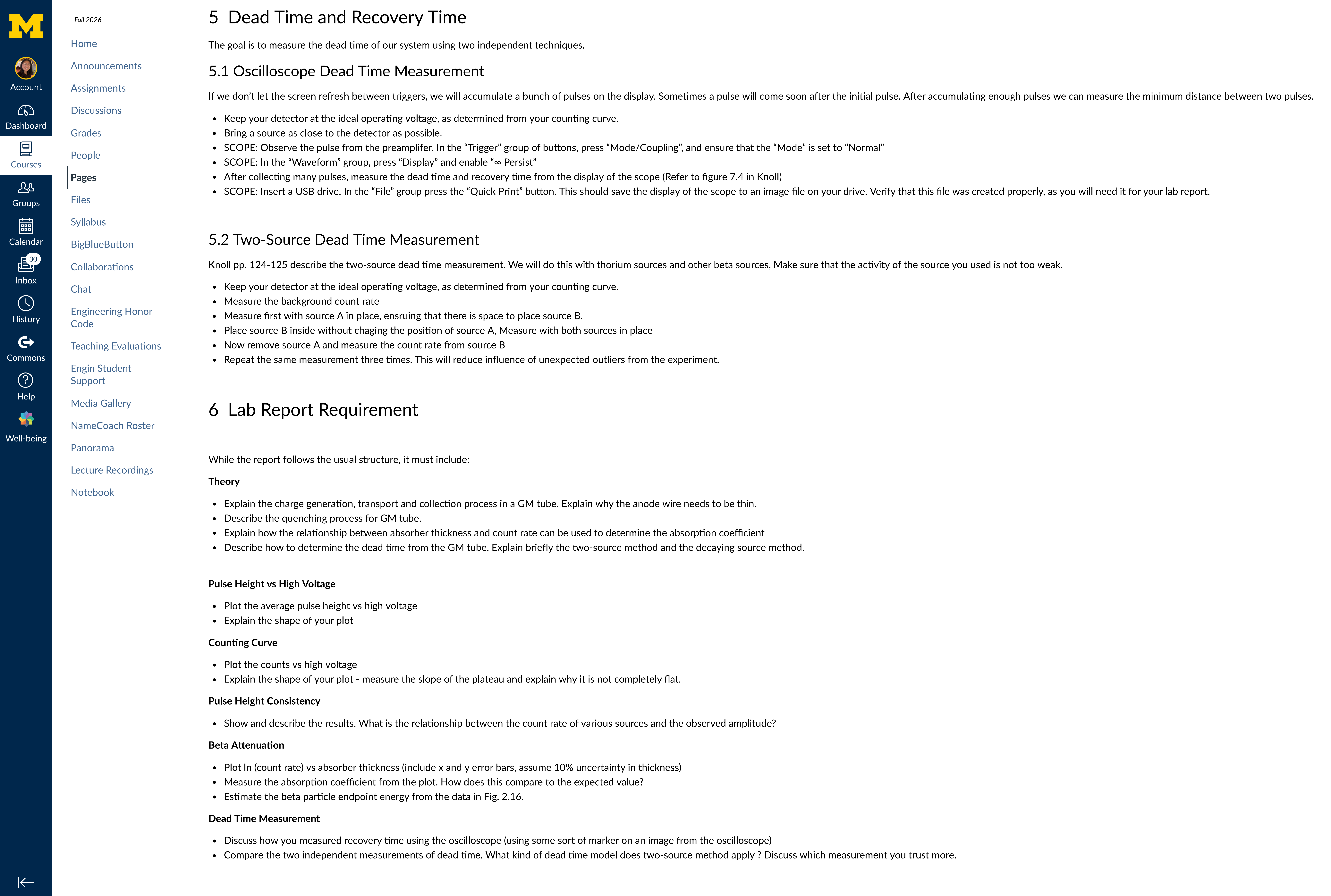

5 Dead Time and Recovery Time

The goal is to measure the dead time of our system using two independent techniques.

5.1 Oscilloscope Dead Time Measurement

If we don't let the screen refresh between triggers, we will accumulate a bunch of pulses on the display. Sometimes a pulse will come soon after the initial pulse. After accumulating enough pulses we can measure the minimum distance between two pulses.

- Keep your detector at the ideal operating voltage, as determined from your counting curve.
- Bring a source as close to the detector as possible.
- SCOPE: Observe the pulse from the preamplifer. In the “Trigger” group of buttons, press “Mode/Coupling”, and ensure that the “Mode” is set to “Normal”
- SCOPE: In the “Waveform” group, press “Display” and enable “∞ Persist”
- After collecting many pulses, measure the dead time and recovery time from the display of the scope (Refer to figure 7.4 in Knoll)
- SCOPE: Insert a USB drive. In the “File” group press the “Quick Print” button. This should save the display of the scope to an image file on your drive. Verify that this file was created properly, as you will need it for your lab report.

5.2 Two-Source Dead Time Measurement

Knoll pp. 124-125 describe the two-source dead time measurement. We will do this with thorium sources and other beta sources, Make sure that the activity of the source you used is not too weak.

- Keep your detector at the ideal operating voltage, as determined from your counting curve.
- Measure the background count rate
- Measure first with source A in place, ensruing that there is space to place source B.
- Place source B inside without chaging the position of source A, Measure with both sources in place
- Now remove source A and measure the count rate from source B
- Repeat the same measurement three times. This will reduce influence of unexpected outliers from the experiment.

6 Lab Report Requirement

While the report follows the usual structure, it must include:

Theory

- Explain the charge generation, transport and collection process in a GM tube. Explain why the anode wire needs to be thin.
- Describe the quenching process for GM tube.
- Explain how the relationship between absorber thickness and count rate can be used to determine the absorption coefficient
- Describe how to determine the dead time from the GM tube. Explain briefly the two-source method and the decaying source method.

Pulse Height vs High Voltage

- Plot the average pulse height vs high voltage
- Explain the shape of your plot

Counting Curve

- Plot the counts vs high voltage
- Explain the shape of your plot - measure the slope of the plateau and explain why it is not completely flat.

Pulse Height Consistency

- Show and describe the results. What is the relationship between the count rate of various sources and the observed amplitude?

Beta Attenuation

- Plot ln (count rate) vs absorber thickness (include x and y error bars, assume 10% uncertainty in thickness)
- Measure the absorption coefficient from the plot. How does this compare to the expected value?
- Estimate the beta particle endpoint energy from the data in Fig. 2.16.

Dead Time Measurement

- Discuss how you measured recovery time using the oscilloscope (using some sort of marker on an image from the oscilloscope)
- Compare the two independent measurements of dead time. What kind of dead time model does two-source method apply ? Discuss which measurement you trust more.